

Hack-Nation · Challenge 01 · “The AI Apprentice” · ElevenLabs

Vashistha

Recorders capture what happened. We learn the decision boundary behind it — and prove it transfers by stopping a mistake on a case the expert never showed us.

11,200 Americans turn 65 daily

Germany 12.9M workers past retirement age by 2036

1.6B people 65+ by 2050

The brief's macro story (as recorded in plan.md §2.3). When a senior reviewer retires, a screen recording keeps the clicks — not the thresholds, exceptions and stop-rules behind them.

LLMs infer. Experts confirm. Code enforces.

LLMs infer

Candidate rules and latent concepts, answer parsing, question phrasing, teach-back prose. Never authoritative.

Experts confirm

A rule is promoted only with the expert's **exact words**, tied to the screen frame they were said over.

Code enforces

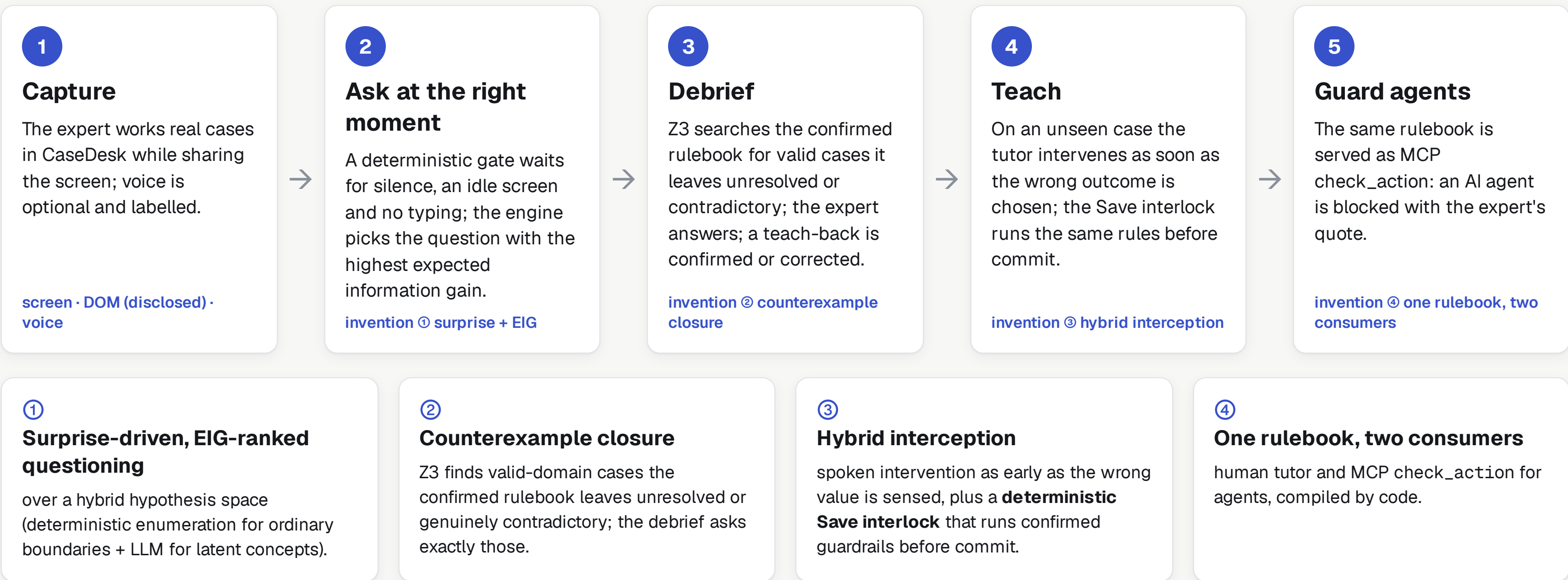
When to speak (gate + nonce), three-valued rule evaluation, Z3 counterexamples, the Save interlock, exports.

What we claim — precisely (plan.md §0)

- **Schema-relative coverage / counterexample closure.** "No unresolved counterexample exists under the current feature model." We **never** claim complete knowledge of the expert. New concepts can always appear; the system is built to absorb them (schema versioning, §6.6).
- **Expected information gain.** computed over an explicit, partly enumerated hypothesis space (§7.3), with a configurable expert-noise heuristic.
- **Deterministic enforcement.** speech authorisation, rule evaluation, Save interlock and exports are code, not model output.

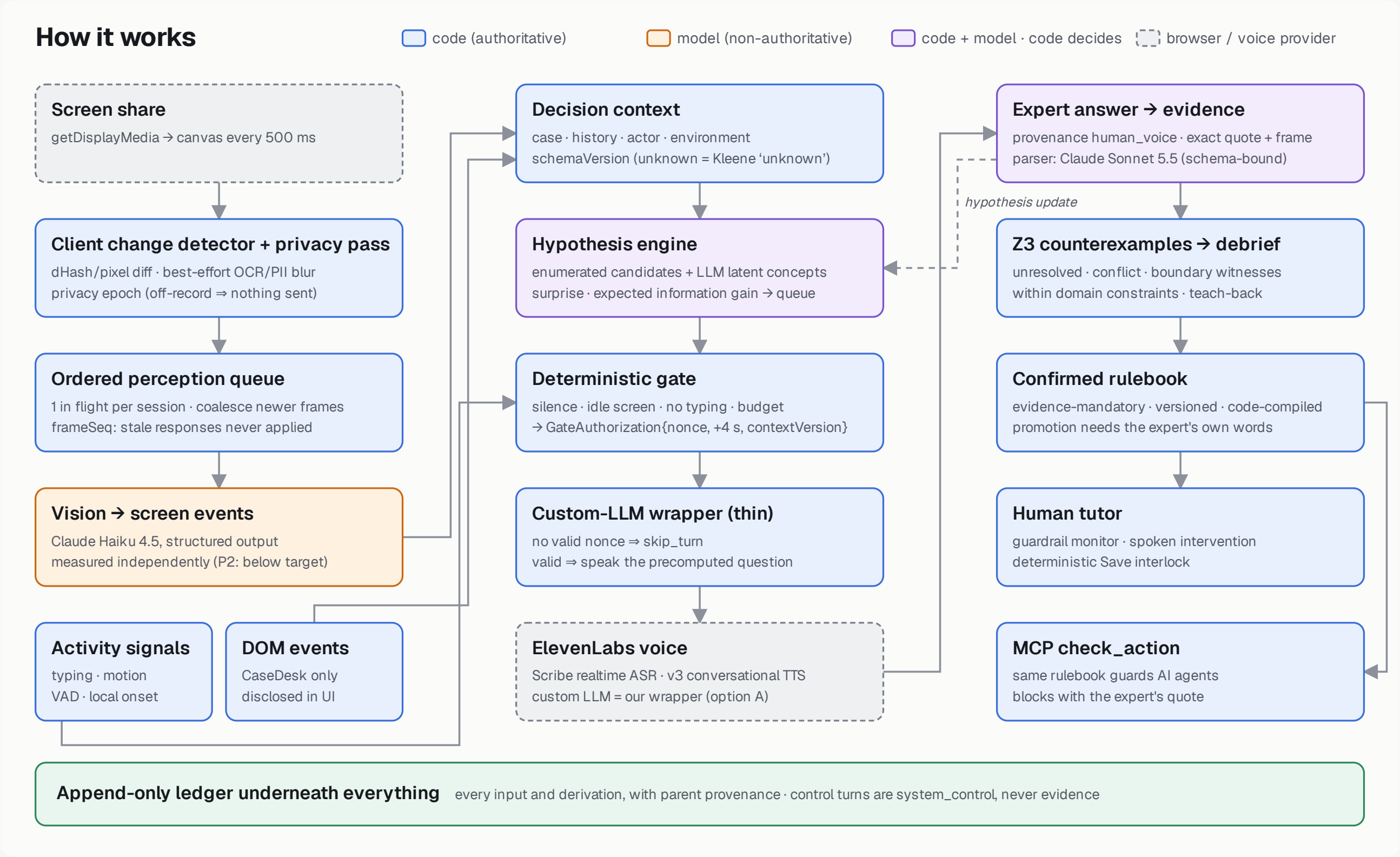
The loop

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Plus **Apprentice-Bench**: a deterministic hidden-policy oracle to measure the loop (plan.md §4, §9).

How it works



Live proof - it asks at the right moment

Northstar Bank

CaseDesk

Synthetic data — fictional policy

Sensing (disclosed): DOM events · typing & scroll timing · microphone only when voice is on

Expert capture

Practice set

e0ca4cd6

+ New session

Case queue

6 of 6 decided

Outcome: Approve onboarding

NS-2026-0302

Decided

Thorncastle Software Studio Holding...

Company · Calderon

Low risk

Outcome: Request documents

NS-2026-0303

Decided

Drummore Freight Partners Holdings ...

Company · Estoria

Medium risk

Outcome: Send to enhanced review

NS-2026-0304

Decided

Bellwether Heritage Trust

Trust · Calderon

Low risk

Outcome: Send to enhanced review

NS-2026-0305

Decided

Eskildsen Aviation Services & Co

NS-2026-0306

Elif Mbatha-Sorensen

Submitted 26 Sept 2026

Customer

NameElif Mbatha-Sorensen

Entity typeIndividual

Registration no. EST-P433934

CountryEstoria

Address140 Old Mill Lane, Valmont, Estoria

Relationship

Customer statusExisting customer

Relationship age8 yr 3 mo

Relationship managerKwame Ashby-Rourke

Beneficial owners

Name	Role	Share	ID verified	PEP
Elif Mbatha-Sorensen	Account holder	100 %	Verified	No

Screening

Source of funds

would you decide?

Expert: If the largest owner holds more than 25% and isn't verified, it goes to enhanced review.

Review

NS-2026-0306

Decision committed

OutcomeApprove onboarding

Risk ratingMedium

InterlockAllowed — no confirmed rule objected

Ledger entry572b8ca0

All DOM events delivered

Screen capture

Capturing

Screen frames: change-detected, best-effort PII

WAITING

Typing ✓

Speaking ✓

Screen moving ⚠ wait 0.9 s

Question value 0.88

Reason: waiting: Screen moving 0.9 s, Budget 438.5 s

Concepts 0

Engineering view

GATE CONDITIONS (ALL 8)

Speaking ✓ 0 ms

Screen moving wait 818 ms

Typing ✓ 0 ms

Breakpoint ✓ 0 ms

Question value ✓ 0 ms

Budget wait 438439 ms

Off record ✓ 0 ms

Agent speaking ✓ 0 ms

Thresholds: silence 1200 ms · screen 1500 ms · typing 1500 ms · 0.3 bits · budget 5/10 min (live questions only: why_probe, counterfactual, concept_definition)

LATENCY · BECAME VALID → AUTHORIZED

p50 274 ms · p95 967 ms · n=5

Bound 250 ms; samples include the authorize round trip. First-audio latency is measured separately.

Keep this tab in the foreground: browsers throttle background timers to ≥ 1 s, which breaks the 250 ms bound.

Gate sensing (timing only, not recorded): keystrokes, scrolling and content changes in the case area; voice activity and agent mode from the voice provider.

QUESTION QUEUE

Context v18 · 5 asked this session

EIG 0.88 bits · counterfactual

"If the relationship age were 0 months instead competing explanations

EIG 0.81 bits · counterfactual

"If the largest beneficial owner share were 45% competing explanations

EIG 0.71 bits · counterfactual

"If the entity type were "company" instead of i competing explanations

EIG 0.50 bits · counterfactual

"If the expected monthly volume were EUR 100 competing explanations

ENGINE STATE

2 confirmed rule(s) · rulebook rev 2

Answer parser: available · unparsed answers 0

reviewOutcome · 6 obs · surprise 0.64 bits

0.17 if customer status is new then enhance

0.06 if entity type is not individual and ana

0.06 if entity type is not individual and cou

0.06 if entity type is not individual and ana

0.02 if source of funds is not not provided

riskRating · 0 obs

0.01 if source of funds is verified then rate

0.01 if largest owner identity verified is no i

Event ticker · live ledger

06:35:04 engine Interlock check · Approve onboard...

06:35:04 dom Decision saved: Approve onboardin...

06:35:04 engine Hypotheses updated · reviewOutcom...

06:35:05 dom Action: Approve onboarding · NS-2...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question queued · competing expla...

06:35:07 engine Question queued · competing expla...

06:35:07 engine Question queued · competing expla...

06:35:07 engine Question queued · competing expla...

Compliance · computed from the ledger

Live Questions 5/3

Guardrail x

Debrief gaps closed 0/3

Teach-back x

Unseen case intercepted x

Production run E: gate conditions, question queue with EIG, live budget · docs/evidence/live/p3/run-e-budget-engineering-view-e0ca4cd6-0101-4cc0-9014-33a0720adbeb.png

3 ms

gate decision p95 after conditions became valid (first live runs)

p50 0 · p95 3 · max 15 ms (n=22)

697 ms

first agent audio p50 after the authorised control message (first live runs)

p50 697 · p95 808 · max 848 ms (n=20)

8 s

silence on an unauthorised turn, end to end through ElevenLabs

preflight 9/9 checks pass · 2026-10-04T03:14:46.513Z

22 authorized · 20 spoken questions in 5 live runs, 2026-10-04 00:27–01:37 UTC.

First live runs: 3 interruptions, all in the noisy-murmur run D → fixes → re-run on the fixed build: 0 interruptions (26 authorized · 26 spoken, 2026-10-04 04:03–04:45 UTC).

Re-run gate decision: 23 of 26 within 250 ms; 3 held or refused authorizations waited up to 5.2 s behind slow production round trips.

Expert speech in these runs: synthetic voice input (ElevenLabs TTS, voice “Matilda”).

Live proof · the expert's words become rules

Promoted live, by voice, from the expert's exact words (P4 met by voice on all four criteria):

“Anything over 25% that isn't verified goes to enhanced review.”

→ `uboOwnershipPct > 25 ∧ ¬uboVerified ⇒ enhanced review`

“Never approve a politically exposed person without compliance sign-off.”

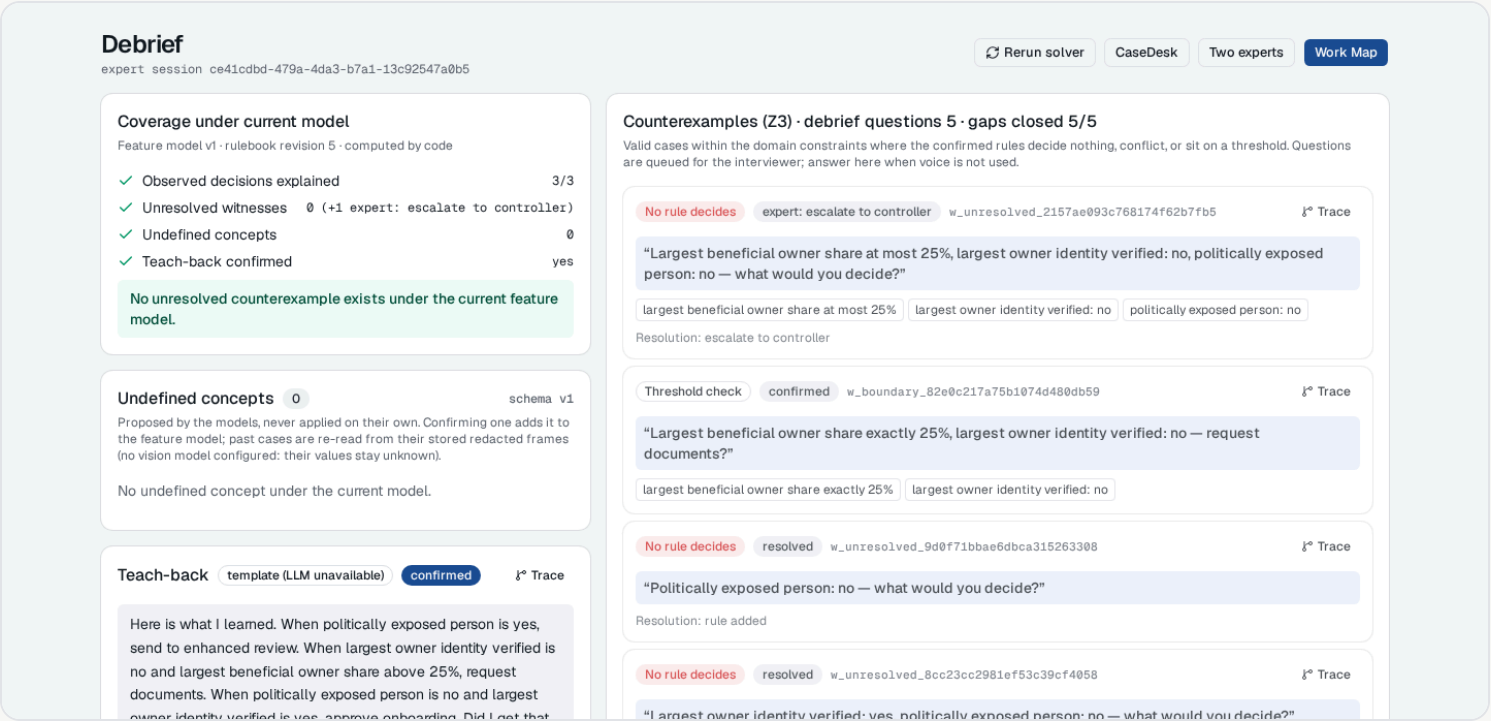
→ `require compliance sign-off when PEP`

“Never approve a customer from a high-risk country at desk level unless they've banked with us for two years with verified funds.”

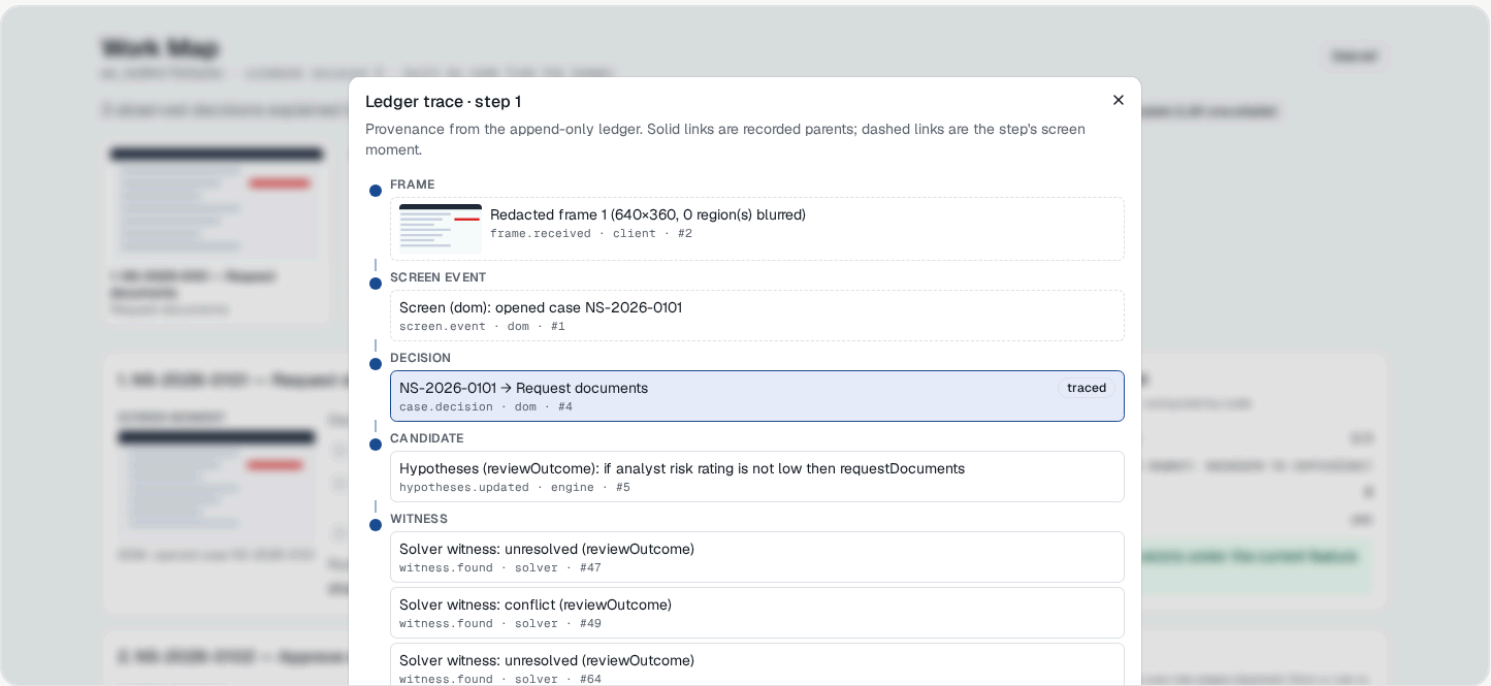
→ `forbid approve when high-risk ∧ < 24 months`

Debrief (e2e): 3 unresolved cells and 1 boundary from Z3 → typed answers → teach-back corrected (rule revised, solver reran) → coverage closed. Live teach-back smoke: Opus, 5.33 s, 57 words.

Coverage closure on the right is the e2e run (typed answers, template teach-back). In the recorded live run the debrief was not completed.



Debrief: coverage under current model, closed · docs/evidence/p5/debrief-coverage-closed.png



Work Map lineage: frame → event → decision → candidate → witness → question → rule · docs/evidence/p5/workmap-lineage.png

Live proof · transfer to a novice and to an agent

Northstar Bank

CaseDesk

Synthetic data — fictional policy

Sensing (disclosed): DOM events · typing & scroll timing · microphone only when voice is on

Novice practice · Held-out set · 77e7a0db

+ New session

Case queue

0 of 2 decided

NS-2026-0201

Marchetti-Sato Textile Imports Ltd

Company · Harrow Bay

High risk

Open

NS-2026-0202

Varga-Holt Commodities Ltd

Company · Dunmore Isles

Medium risk

Open

NS-2026-0201

Marchetti-Sato Textile Imports Ltd

Submitted 29 Sept 2026

Customer

Name

Marchetti-Sato Textile Imports Ltd

Entity type

Company

Registration no.

HAR-C330981

Country

Harrow Bay

High risk

Address

88 Harbour Parade, Gullhaven, Harrow Bay

Relationship

Customer status

New customer

Relationship age

New relationship

Relationship manager

Rafael Moreno-Teague

Beneficial owners

Name	Role	Share	ID verified	PEP
Chiara Marchetti-Sato	Director & shareholder	30 %	Verified	No
Yusuf Ellingham	Shareholder	25 %	Verified	No
Greta Featherstone	Shareholder	15 %	Verified	No

Screening

Sanctions

Clear

No match on the Northstar Synthetic Sanctions List.

Adverse media

None

Source of funds

Status

Verified

Description

Trading revenue; audited 2025 accounts and bank statements reviewed.

Review

NS-2026-0201

Risk rating

high-risk country always goes to enhanced review."

The expert, typed during the debrief

Replay the expert's moment

Careful — the expert's rule forbids "Approve onboarding" here

Queued for the tutor's voice: spoken as soon as the agent is idle: Careful — never approve onboarding when country risk is high. The expert said: "Never approve a customer on a high-risk country list at desk level."

When country risk is high: never approve onboarding

"Never approve a customer on a high-risk country list at desk level."

The expert, typed during the debrief

Replay the expert's moment

Save runs the same check: this outcome will be blocked.

LISTENING

Typing ✓

Speaking ✓

Screen moving ⌘ wait 1.1 s

Question value

Reason: no question queued

Engineering view

Event ticker · live ledger

08:37:46 engine Session started · novice practice · heldout set

Compliance · computed from the ledger

Live questions 0/3

Guardrail x

Debrief gaps closed 0/3

Teach-back x

Unseen case intercepted x

Unseen case NS-2026-0201: the tutor intervenes on selection, before Save · docs/evidence/p6/intervention-card-before-save.png

Real Claude (claude-opus-5-5) → production /mcp · 2026-10-04T01:28:06Z

→ check_action {"proposedAction":"approve", case NS-2026-0201 ...}

← forbid (rulebook revision 23) by rule_1e5796de1f0929

BLOCKED: "Never approve a customer from a high-risk country at desk level unless they've banked with us for two years with verified funds."

132/132

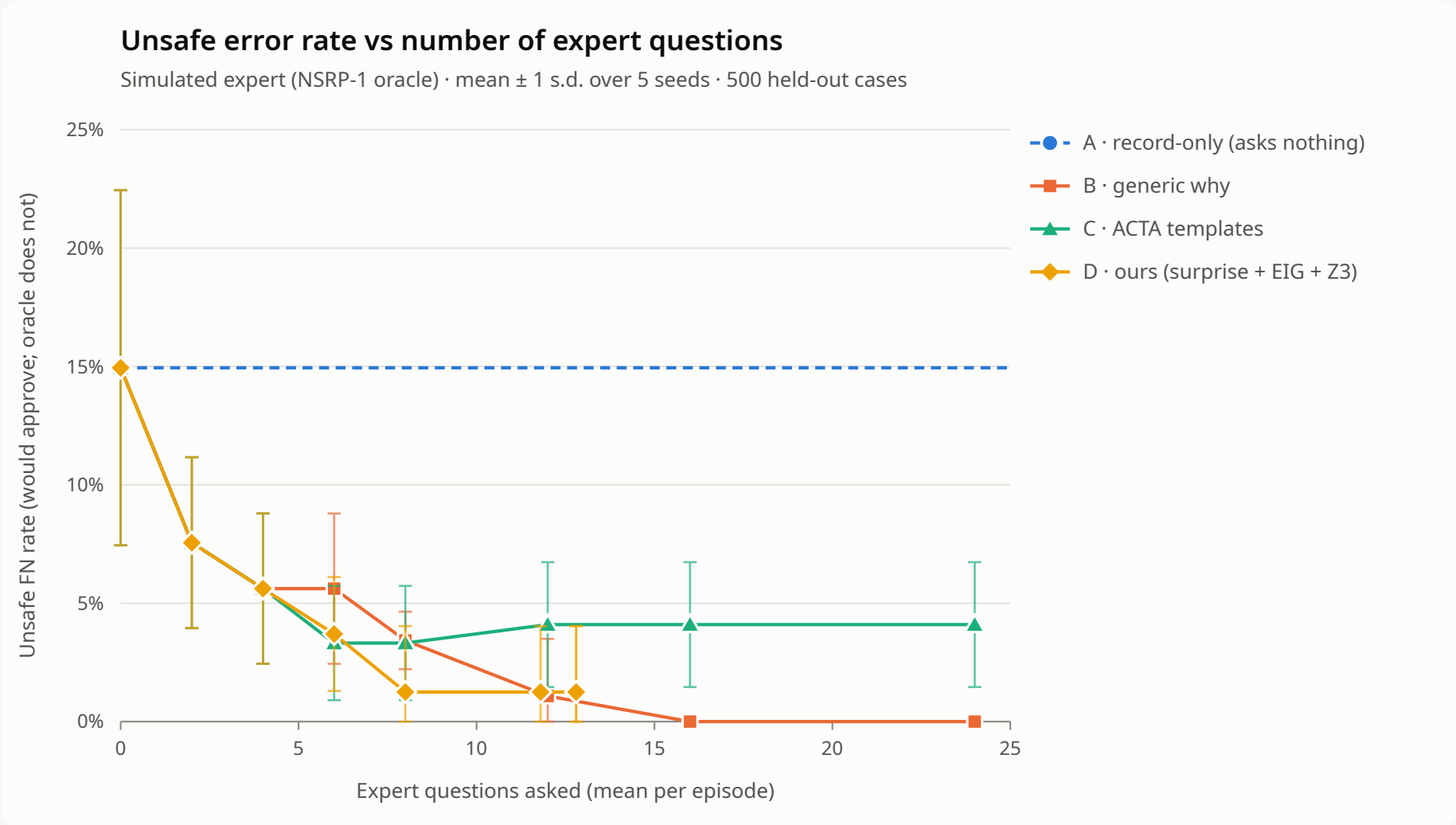
violating commits blocked by the Save interlock (property test, end to end)

55/55

MCP check_action agrees with the local interlock (11 demo cases × 5 outcomes, production)

The quote the agent receives is the production utterance from the live capture run.

Apprentice-Bench · the money chart



docs/evidence/bench/unsafe-vs-questions.svg

14.9% → 1.2%

unsafe approvals: record-only floor → ours (D) after 8 questions
fidelity 0.633 → 0.990

! D vs B, honestly

Generic “why” after every step (B) reaches 0.0% unsafe — but needs 16–24 questions and as many interruptions. D plateaus at 12.8 questions (7.8 interruptions) with 1.2% unsafe and the best fidelity (0.990 vs 0.949).

B is safer than D at budgets ≥ 12. On 1/5 seeds D never learns the adverse-media rule (it only co-fires with the high-risk rule): the schema-relative limit we state. With 2 questions guardrail recall dips below the floor (94.1% vs 98.3%).

Simulated expert: a deterministic oracle (NSRP-1, fictional) answers; 5 seeds × 500 held-out cases. Not a human-subject result.

Trust, privacy and verified replay

Off the record

Mutes the mic first, stops frame capture, cancels queued uploads and advances the privacy epoch; the server refuses stale uploads (409).

“Off-record content is prevented from entering our evidence store; microphone and frame transmission are disabled while off the record.”

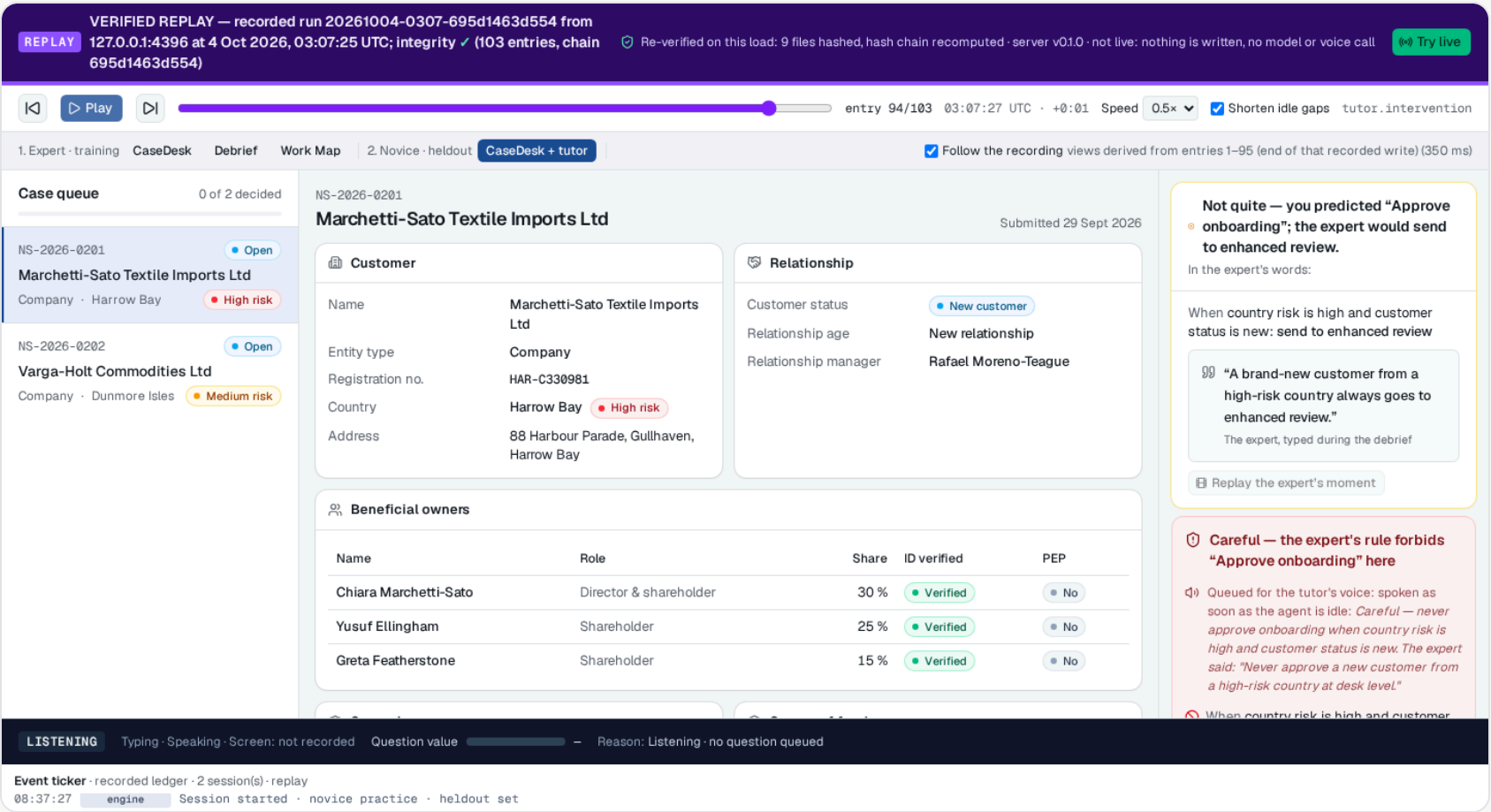
The trigger phrase itself may reach the voice provider; only a marker is stored, never the words. Our agents set 30-day retention.

PII and data

Best-effort OCR blur on the client before upload. The sandbox is fully synthetic (fictional bank, people, countries) — that is the real privacy guarantee for the demo. Every rule needs expert sign-off; evidence is the expert's own words.

Verified replay of a recorded run

Bundle 20261004-0119-96acd563eb14: 393 ledger entries from 2 live sessions on vashistha-production.up.railway.app, 2 recorded conversations. sha256 per file + hash chain (head 96acd563eb14...), re-verified on every load; one changed byte ⇒ refused. Same UI; nothing is written, no model or voice call.



Verified replay UI (e2e bundle): violet banner, integrity ✓, the live components · docs/evidence/p11/replay-intervention.png

Stretch · two experts and Hindi → English

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Two experts

Where two experts' confirmed rules decide the same valid case differently — and how they settled it.

First expert
Second expert
Decision
Find disagreements (Z3)

Asha Rao's rulebook · Review outcome

decision held back
rule_1018929f76caec · r2 · p10

When country risk is high → **send to enhanced review**

"Anything from a high-risk country goes to enhanced review."

decision held back
rule_0b64f772327077 · r2 · p30

When customer status is existing and relationship age at least 24 months and country risk is high → **approve onboarding**

overrides rule_1018929f76caec

"Unless they have banked with us for two years or more; then I approve."

Priya Sharma's rulebook · Review outcome

decision held back
rule_09c3a060326266 · r2 · p10

When country risk is high → **send to enhanced review**

"High-risk country means enhanced review, every time."

guardrail
rule_13e28f9a79d553 · r1 · p40

When country risk is high → **never approve onboarding**

"अगर देश हाई-रिस्क लिस्ट पर है तो मैं डेस्क लेवल पर अप्रूव नहीं करती।"

A valid case where they decide differently asked both experts
w_disagreement_7bfda69638d9c8c16cc143e9

Found by Z3 over both confirmed rulebooks, within the domain constraints. When found: Asha Rao → Approve onboarding, Priya Sharma → Send to enhanced review.

Customer status	existing
Relationship age (months)	24 months
Country risk (Northstar list)	high

Asha Rao

Their rulebook decides: **Approve onboarding**

Question for Asha Rao (queued): "Customer status existing, relationship age at least 24 months, country risk high — approve onboarding or send to enhanced review?"

No answer yet.

Asha Rao's decision

Approve onboarding
>

Your words (recorded as evidence)

Record Asha Rao's decision

Priya Sharma

Their rulebook decides: **Send to enhanced review**

Question for Priya Sharma (queued): "Customer status existing, relationship age at least 24 months, country risk high — approve onboarding or send to enhanced review?"

No answer yet.

Priya Sharma's decision

Approve onboarding
>

Your words (recorded as evidence)

Record Priya Sharma's decision

Team rulebook in force · Review outcome · 1 open disagreement

What the Save interlock, the tutor and MCP check_action enforce: every expert's confirmed rules, except decision rules held back (marked) while their experts disagree on a case. Guardrails are never held back, so a disagreement can only make the interlock stricter.

decision held back Asha Rao
rule_1018929f76caec · r2 · p10

When country risk is high → **send to enhanced review**

"Anything from a high-risk country goes to enhanced review."

Two experts

Z3 searches both rulebooks for a valid case where they decide differently; each expert is asked; the resolution is a rule revision carrying **both** quotes. While open, only decision rules are held back — a forbid from either expert always applies (monotonic safety, property-tested).

Hindi → English

Language detection is code; Sonnet translates; code checks the translation segments are verbatim and complete. Quotes are stored in the original with a labelled machine translation; the tutor speaks English.

✓ Live Hindi run on production: end to end Synthetic Hindi voice →

ElevenLabs ASR → verified translation → Hindi-quoted rule → MCP block
(docs/evidence/live/rerun-2026-10-04/ACCEPTANCE.txt).

Two experts: Z3 finds a valid case where their rulebooks disagree · docs/evidence/p10/two-experts-disagreement.png

Where we sit

	Captures steps	Elicits reasons	At the screen moment	Counterexamples verified by expert	Coaches humans on unseen cases	Guards agents
Recorders (Scribe/Tango/Guidde)	✓	—	✓	—	partial	data only
Skan AI	✓	inferred	from traces	—	—	✓
Tacit / KNOA	—	✓	—	claim checks	—	—
DAPs (WalkMe/Whatfix/Userlane)	—	—	✓	—	authored	—
Policy guardrails (AWS/Cedar)	—	—	—	on given policy	—	✓
Vashistha	✓	✓	✓	✓	✓	✓

Adjacent categories are real and funded: Scribe \$75M Series C at \$1.3B valuation (company press release); SAP acquired WalkMe for ~\$1.5B; Skan AI \$63M Series C (Aug 2026). We explore a different acquisition mechanism: active, screen-grounded interrogation of a live decision, counterexamples verified by the expert, and transfer tests.

What we measured — and what we missed

Acceptance	Result	Detail
P2 vision (live, Haiku 4.5)	✗ 0/4 thresholds met	critical field-change recall 0.192 → 0.577 (target ≥ 0.95) · critical action recall 0.262 → 0.667 (target ≥ 0.95) · false critical rate 0.432 → 0.227 (target ≤ 0.05) · p95 frame→event 8.9 s → 3.7 s (target ≤ 3 s). Tutor and interlock use the disclosed DOM channel; team decision on options pending.
P3 zero interruptions (5 live runs)	✓ 0 in the re-run	First live runs: 3, all in run D (quiet murmurs below the VAD threshold; a control message swallowed by an open user turn). Fixed (local onset detector, open-turn tracking, re-queue); re-run on the fixed build: 0 across 5 runs, 26 authorized · 26 spoken.
P3 authorise ≤ 250 ms	! 23 of 26 in the re-run	First runs: p95 3 ms. Re-run: p50 0 · p95 4157 · max 5228 ms (n=26) — 3 held or refused authorizations behind slow production round trips. Not met strictly.
P4 hypothesis engine, live by voice	✓ met	Threshold rule, PEP guardrail, unresolved concept; every promoted rule passes evidence validation. Attempts 1–3 kept, with the bugs they found.
P5 counterexample closure	! met (e2e, typed)	Live voice debrief not yet run end to end.
P6 tutor + Save interlock	✓ 132/132 blocked	Intervention ledgered before Save; unseen case handled; Z3 practice cases.
P8 agent + exports (live)	✓ 55/55 agree	Real Claude blocked with the expert's quote; Work Map/Procedure round-trip byte-identical.
P10 two experts + Hindi live	✓ done	Two experts in e2e; Hindi → English end to end on production (synthetic Hindi voice, real ASR, verified translation, Hindi-quoted rule, MCP block).
Event loop (bug #6)	! p99 1.8 ms · max 13.9 s	Before (production): /api/health stalled 10.5 s and 5.3 s during a cold Work Map export; handler gaps up to 4.6 s. Worker threads landed in 1b0c089; production preflight 2026-10-04T03:14:46.513Z: event-loop delay p99 2 ms; repo heavy-flow test on 64× INTEL(R) XEON(R) GOLD 6530: worst block 39.3 ms. After the live re-runs production showed one 13.9 s stall at p99 1.8 ms: diagnosed as a host-level freeze; mitigations listed, not yet verified.

Numbers from docs/evidence/p2/eval-live-4.json, docs/evidence/live/{ACCEPTANCE,SUMMARY}.txt, docs/evidence/live/bugs/*, PROGRESS.md, docs/video/evidence/event-loop-local.json. Thresholds were never moved to pass.

NOW

One expert → verified rulebook → tutor + agent guardrail.

NEXT

Company Memory: every expert, one rulebook that stays current; when work changes, ask only about what's new.

THEN

Always-on apprentice: notices never-seen cases during normal work.

MOONSHOT

The verified organisational judgment layer: every human and every agent decides routine cases from the same expert-approved rulebook — audit-ready.

LLMs infer. Experts confirm. Code enforces.